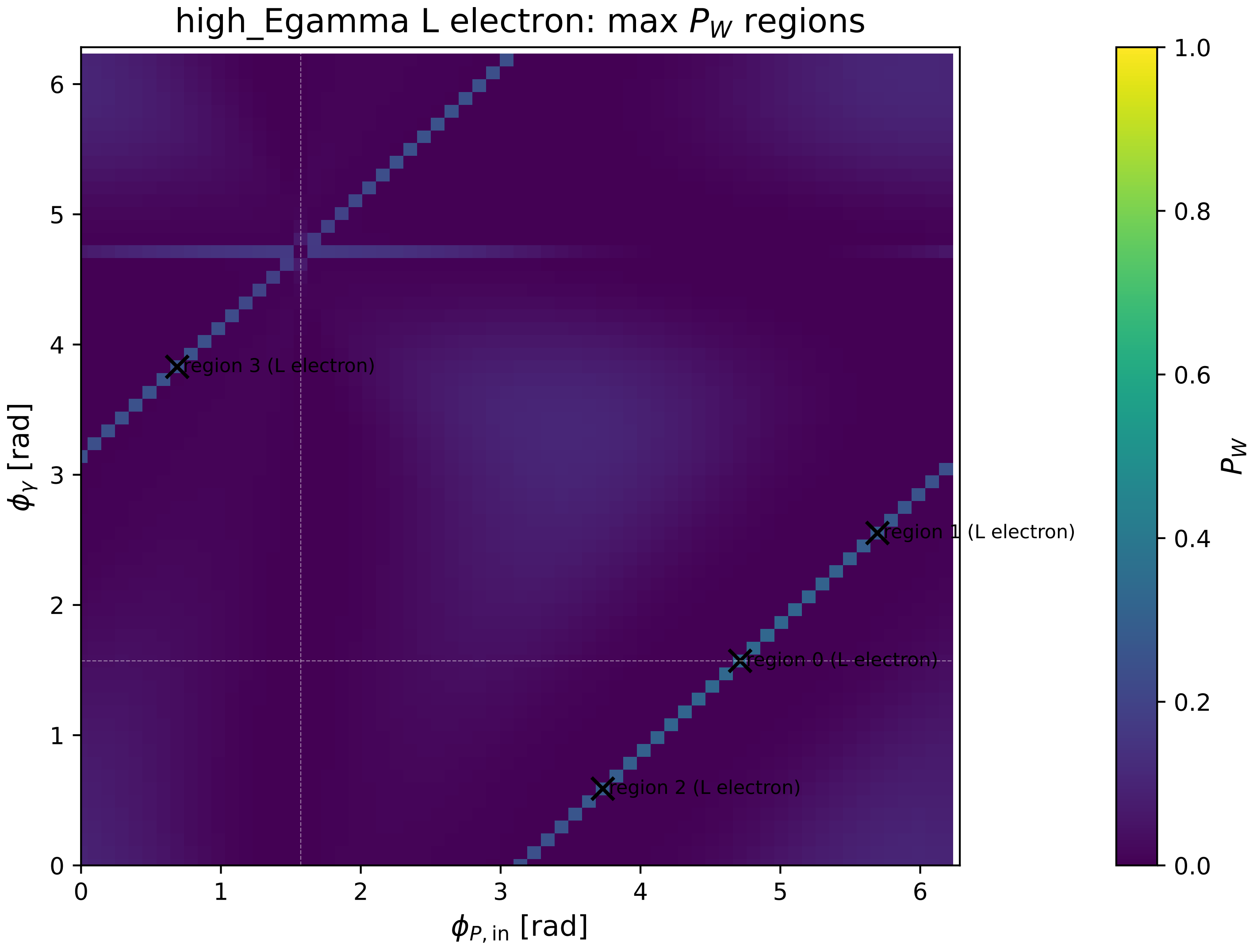
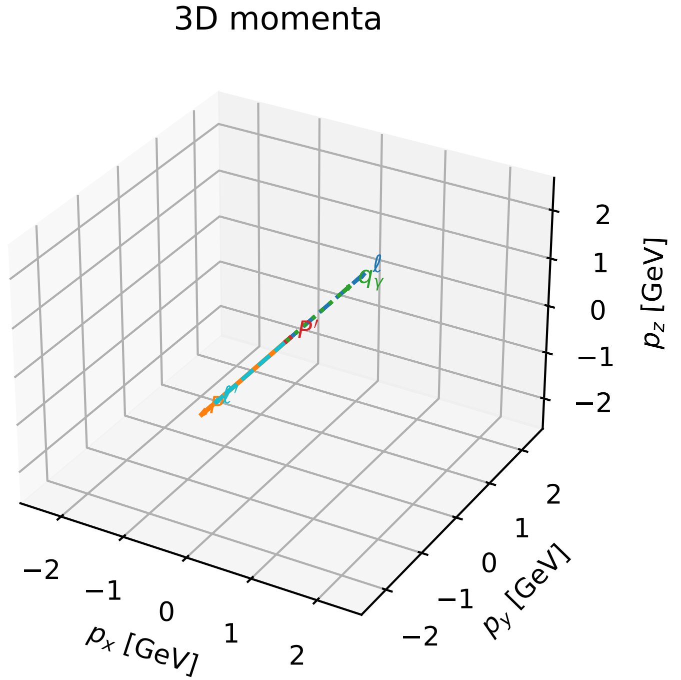


high_Egamma L electron: max P_W regions
region 3 (L electron)
region 1 (L electron)
region 0 (L electron)
region 2 (L electron)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

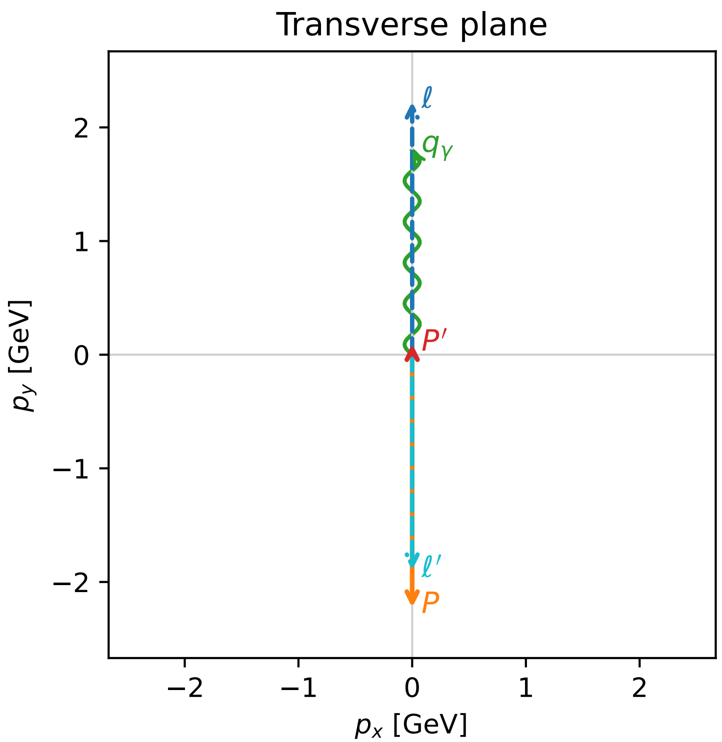
L electron characteristic max P_W configuration

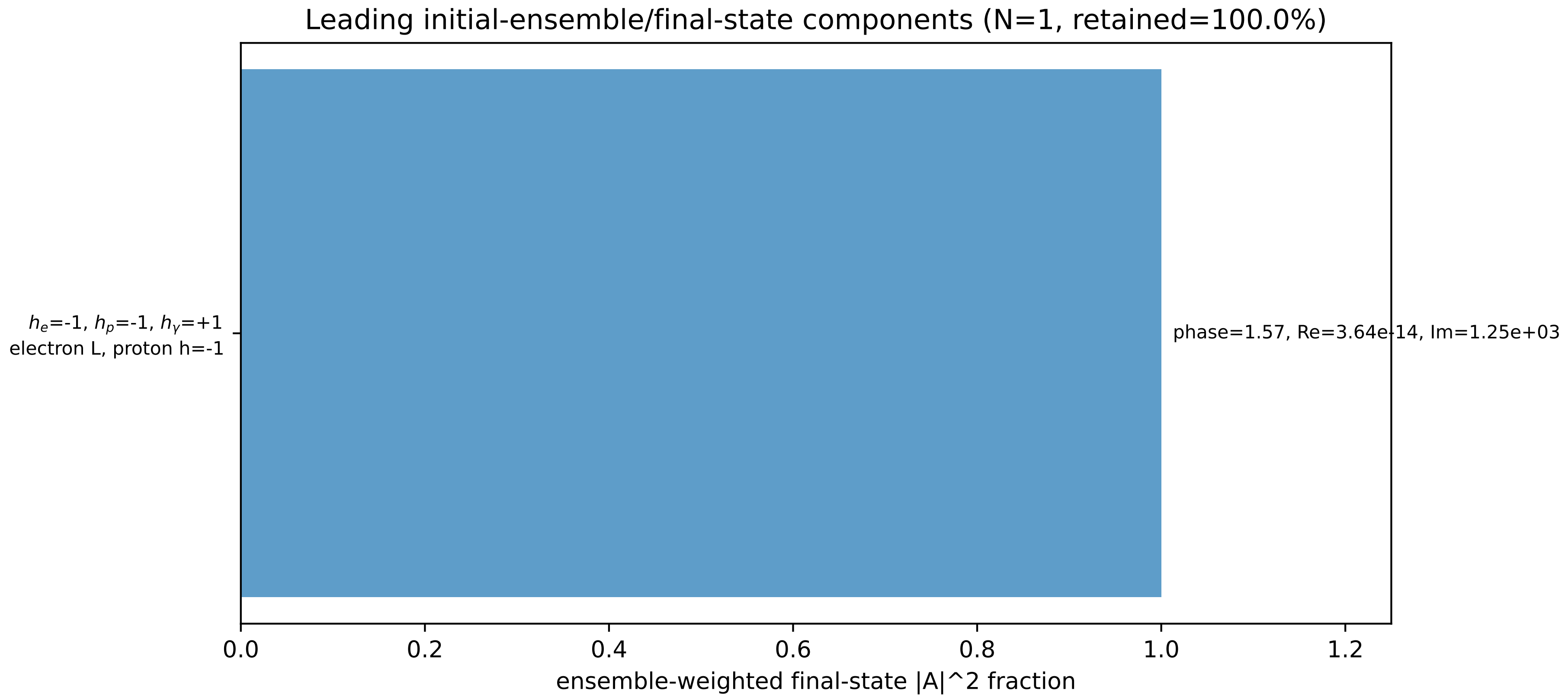


w_purity_high_egamma_l_lepton_region_0 P_W=0.333316
region: high_Egamma_L_lepton_region_0
outgoing-state purity: 0.999894
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

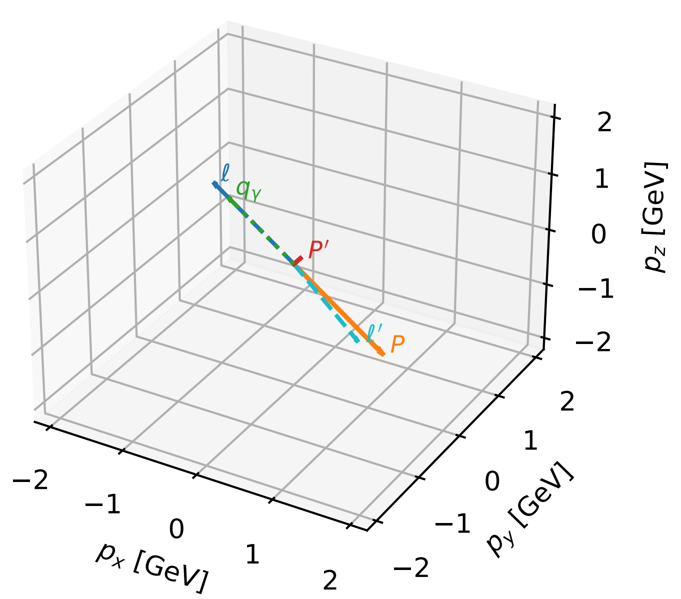
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0956529
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=1.5708
Q^2=16.8669, x_B=0.846865, t=-3.21819, W^2=3.92981, y=0.965151
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.8957 p_x= -0.0000 p_y= -1.8957 p_z= 0.0000
P' E= 0.9429 p_x= 0.0000 p_y= 0.0957 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.0000 p_y= 1.8000 p_z= 0.0000





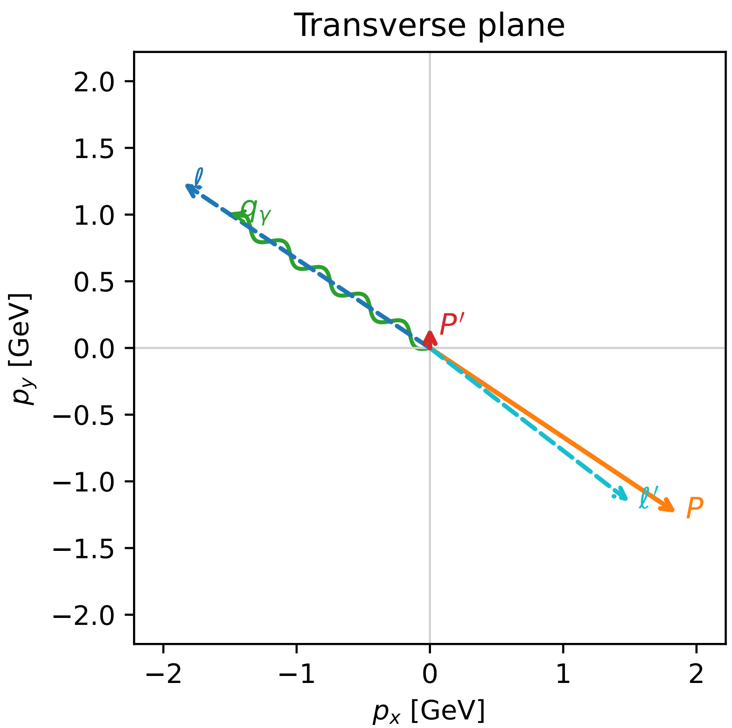
L electron characteristic max P_W configuration

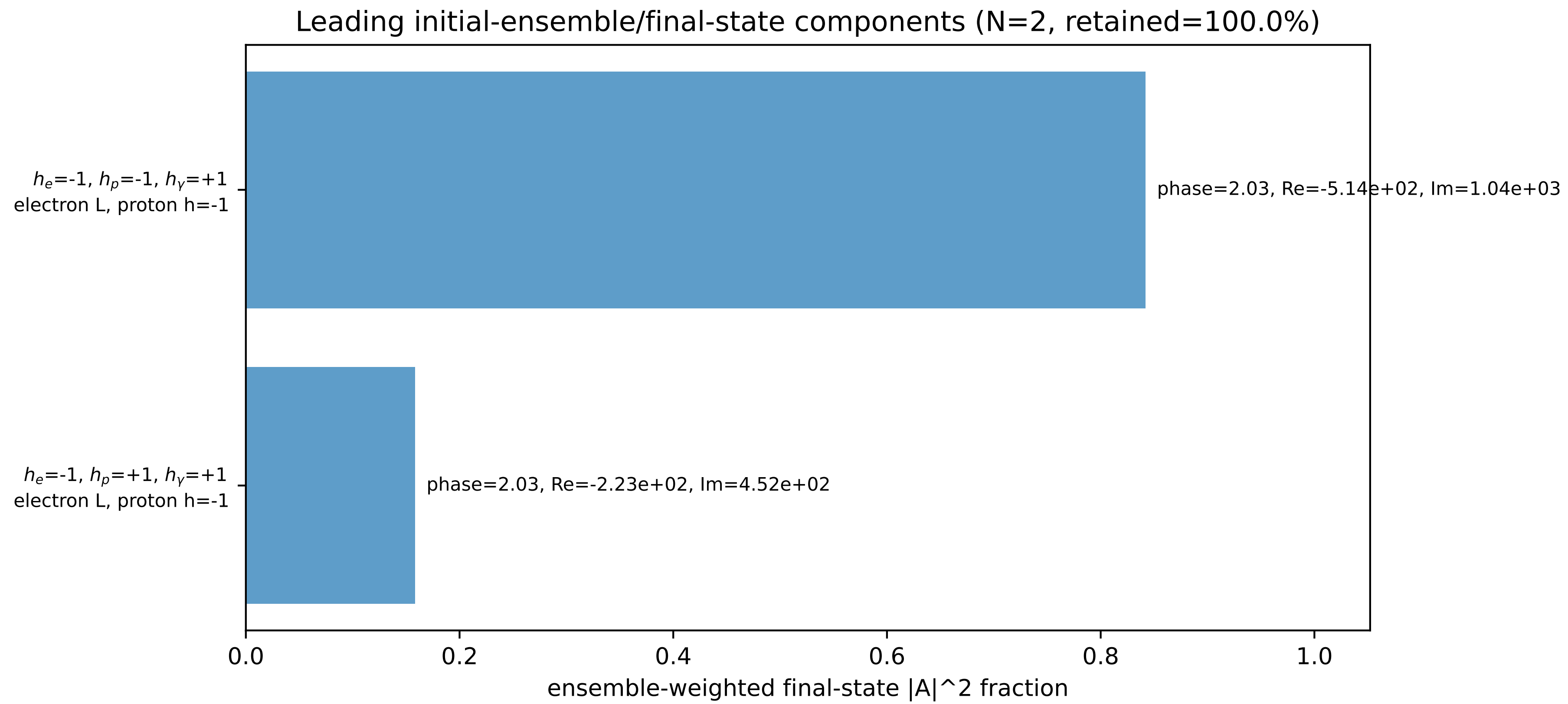


w_purity_high_egamma_l_lepton_region_1 P_W=0.280629
region: high_Egamma_L_lepton_region_1
outgoing-state purity: 0.999867
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.151482
θ_{in}=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=1.8, ϕ_γ=2.55254
Q^2=16.7833, x_B=0.843345, t=-3.20225, W^2=3.99742, y=0.964378
$\theta(\ell', \gamma)$=176.176 deg

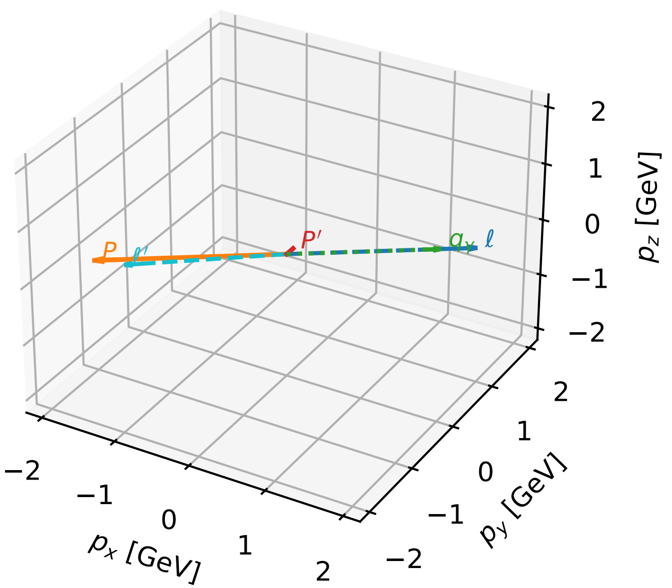
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= 1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 1.8884 p_x= 1.4966 p_y= -1.1515 p_z= 0.0000
P' E= 0.9502 p_x= 0.0000 p_y= 0.1515 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.4966 p_y= 1.0000 p_z= 0.0000





L electron characteristic max P_W configuration

3D momenta



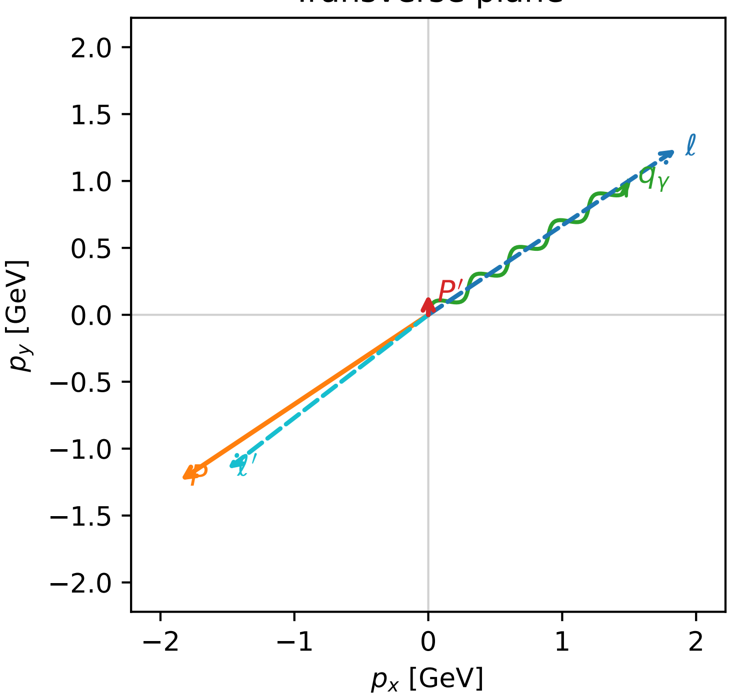

```
w_purity_high_egamma_l_lepton_region_2  P_W=0.280623
region: high_Egamma_L_lepton_region_2
outgoing-state purity: 0.999867
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

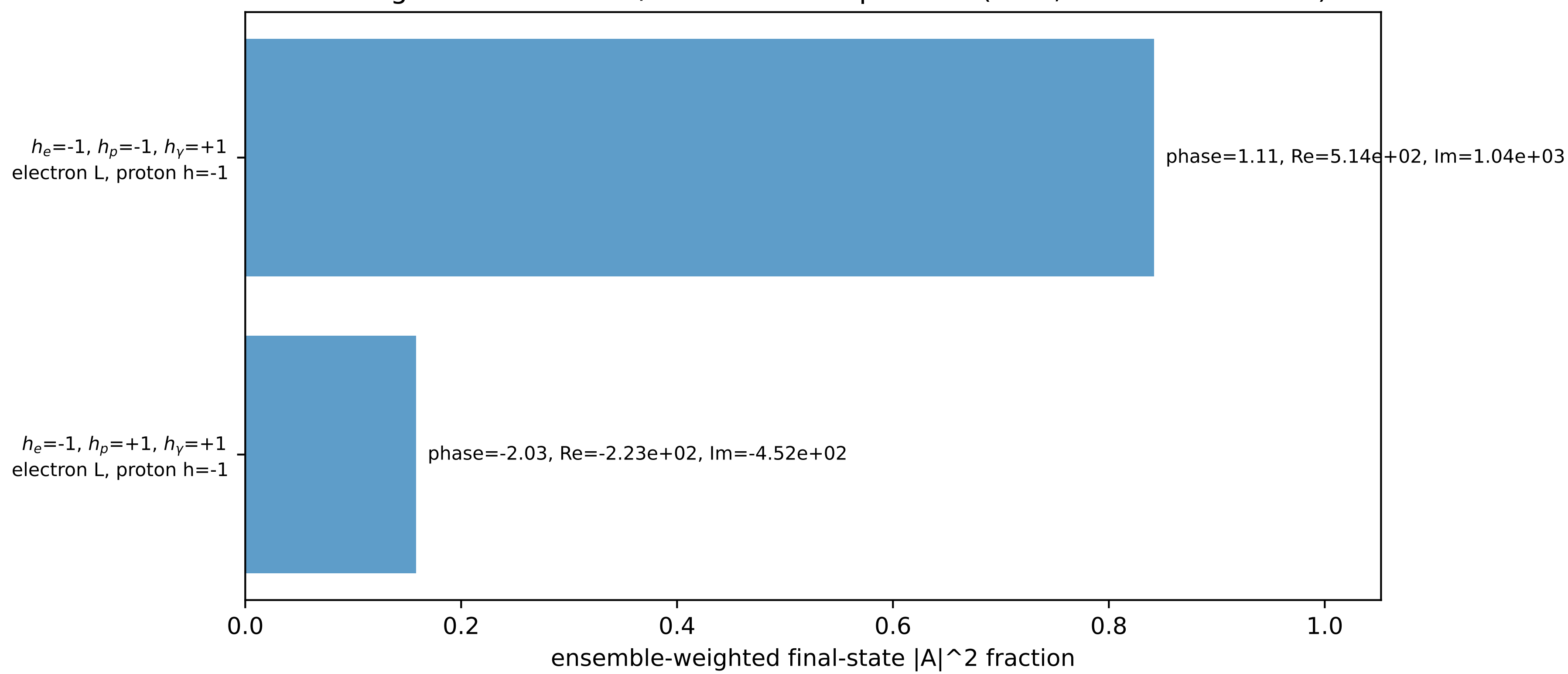
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.151482
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=1.8, ϕ_γ=0.589049
Q^2=16.7833, x_B=0.843345, t=-3.20225, W^2=3.99742, y=0.964378
$\theta(\ell', \gamma)$=176.176 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= -1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 1.8884 p_x= -1.4966 p_y= -1.1515 p_z= 0.0000
P' E= 0.9502 p_x= 0.0000 p_y= 0.1515 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.4966 p_y= 1.0000 p_z= 0.0000

Transverse plane

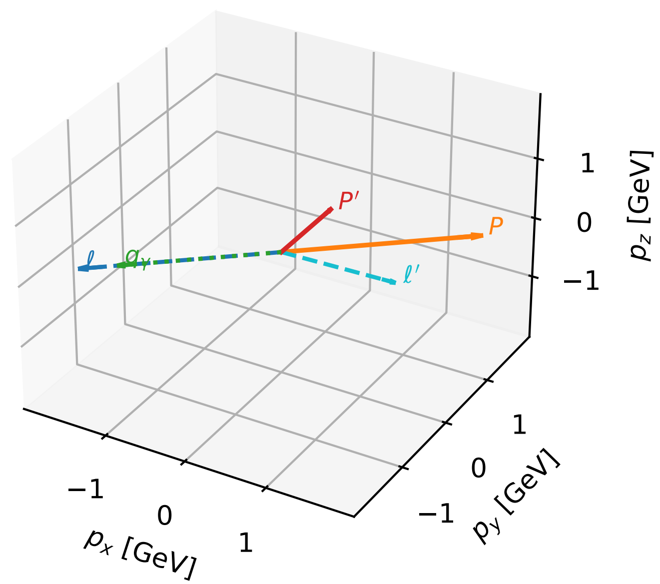


Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L electron characteristic max P_W configuration

3D momenta




```
w_purity_high_egamma_l_lepton_region_3  P_W=0.251919
region: high_Egamma_L_lepton_region_3
outgoing-state purity: 0.876053
kinematic point: high_s_high_theta_in_high_Egamma
```

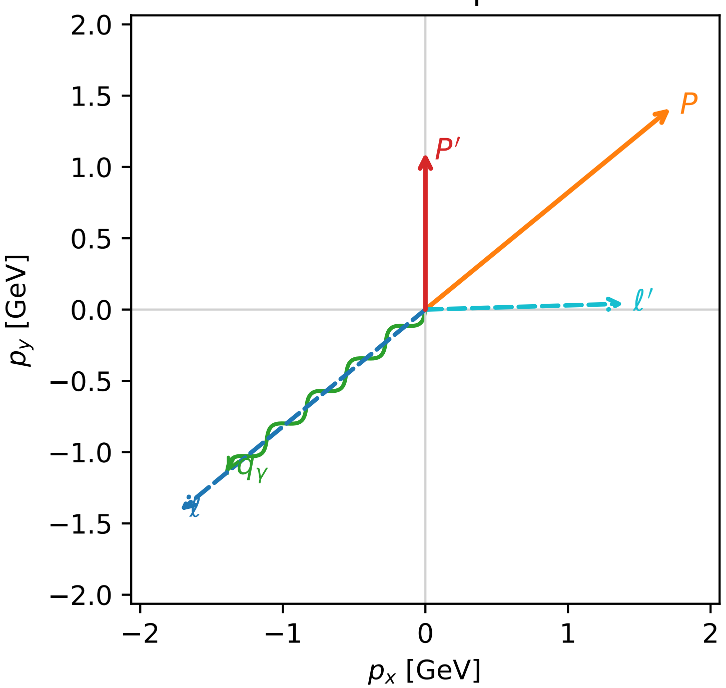
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.10115
$\theta_{\rm in}$=1.57079, ϕ_ℓ=3.82882, ϕ_P=0.687223
E_γ=1.8, ϕ_γ=3.82882
Q^2=11.093, x_B=0.589575, t=-2.11653, W^2=8.60206, y=0.911766
$\theta(\ell', \gamma)$=142.303 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= -1.7195	p_y= -1.4112	p_z= -0.0000
P	E= 2.4141	p_x= 1.7195	p_y= 1.4112	p_z= 0.0000
ℓ'	E= 1.3920	p_x= 1.3914	p_y= 0.0408	p_z= 0.0000
P'	E= 1.4465	p_x= 0.0000	p_y= 1.1011	p_z= 0.0000
q_γ	E= 1.8000	p_x= -1.3914	p_y= -1.1419	p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)